# COGNI TRACE : JOB SEARCH AND RECRUITMENT PLATFORM

A Course Project Report

Submitted in the partial fulfillment of the completion of the course

**25CAI302SC507- Human-Computer Interaction and Design**

**In I Year II Semester M.Tech- CSE**

**POST GRADUATION**

in

**SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE**

by


| | |
|---|---|
| **CHITTIMALLA PUJITHA** | **(2503B05118)** |
| **MANNE EVANCY** | **(2503B05116)** |
| **KATAKAM SHREEMAYI** | **(2503B05124)** |


Under the guidance of

**Dr. Bediga Sharan**

Assistant Professor, School of CS&AI.

SR UNIVERSITY

SR University, Ananthsagar,Warangal,Telagnana-506371

# TABLE OF CONTENTS


| S. No. | Content Title | Page No. |
|---|---|---|
| 1 | Abstract | 1 |
| 2 | About the Organization | 2 |
| 3 | Introduction of Project | 3 - 6 |
| 4 | Related Work | 7 - 11 |
| 5 | Problem Statement | 12 - 17 |
| 6 | Requirement Analysis, Risk Analysis, Feasibility Analysis | 18 - 27 |
| 7 | Proposed Solution | 28 – 34 |
| 8 | Architecture Diagram, Flow Chart | 35 – 36 |
| 9 | Simulation Set up and Implementation | 37 - 42 |
| 10 | Result Analysis | 43 - 50 |
| 11 | Learning Outcomes | 51 - 52 |
| 12 | Conclusion with Challenges | 53 |
| 13 | Future Scope | 54 - 55 |
| 14 | References | 56 |

# ABSTRACT

CogniTrace is a web application that is modern and user favored and developed to simplify the job seeking and application process of students and early-career professionals. The frontend is a web application based on ReactJS, and the back end is Django. An active user interface is connected with a stable and effective server-side platform. It has an OTP authentication system, which uses email to create safe and verified access. The users will be provided with a responsive and intuitive home page displaying a broad range of jobs that are in accordance with their skills and interests after verification. The platform is well designed and has effective navigation that enables users to view job listings, add resumes and apply directly in the platform. With a new perspective in the alternative to job portals guided by recruiters, CogniTrace will provide the user with a greater sense of control over their career path, offering transparency, relatability, and ease. In this project, the real-life application of a full-stack development is presented through knitting together hi-fi web technologies into a smooth and secure user experience that is exciting.

# 1. ABOUT THE ORGANISATION

The project was developed at the School of Computer Science and Artificial Intelligence, SR University, Warangal which is one of the topmost innovation oriented institutions in the country. SR University is geared towards experiential learning, research-based learning, and solving real-world problems, therefore, empowering students to design and implement effective technological solutions.

One of the pillars in training its students to get them ready to work in the industry is the Course Project Program at SR University. Under this program, students are required to work as a team in the development of end-to-end systems addressing real world problems with the most up to date technologies. Every project is evolved through guidance of the mentored faculty and constant guidance and technological support.

This academic project led to the development of CogniTrace. CogniTrace is a job search and job recruitment site that is student-focused and that is created with ReactJS, Django REST Framework and SQLite3, with the perspective of simplifying career paths of new graduates and young professionals. The system concentrates on safe authentication, customized job searching, resume handling, skill monitoring and interview preparation applications to fill the disparity between student abilities and industry anticipations.

# 2. INTRODUCTION OF PROJECT

## 2.1 OVERVIEW:

One of the most crucial stages in the career of a student is finding a job, although, it may be confusing, overwhelming, and extremely competitive. The current job portals are constructed with the main purposes of the recruiters rather than the candidates. They use generalized key word search, general filters and sophisticated interfaces which cause difficulty to the students in finding opportunities that match their real skills, interests and career objectives. Because of that, recent graduates often find themselves on endless lists without knowing what position suits them and how to prepare.

CogniTrace was created to fill this void by developing a student-centrist, no-frills, recruitment platform that is personalized, easy to understand, use and follow. CogniTrace will not flood the user with useless feeds on available job opportunities but will match the skills, qualifications, and preferences with the appropriate opportunities. It allows students to build a professional digital profile, post resumes, browse curated jobs, monitor applications, and ready to interview-All in a clear, easy to navigate interface.

CogniTrace is a modern web application that is built on the top of ReactJS and Django REST Framework, which guarantees efficiency, speed, and security. The application will also have basic tools like resume management, skill tracking, job recommendations, and interview preparation tools. CogniTrace can also become a great guide to students who are on their initial experiences with job search websites and make the process more organized, transparent, and confidence-building.

CogniTrace is more than a job portal it is a step towards candidate-centric technology of recruitment a platform that is sensitive to the challenges encountered by the student and that assists them with success enabling tools.

## 2.2 OBJECTIVES OF THE PROJECT:

CogniTrace is mainly aimed at building a safe, easy to use job search engine that will enable students to find jobs easily and have the ability to control their careers without any fear. The system is a combination of job filtering, profile making, resume posting, and personalized recommendations.

**The targeted objectives are:**

1. To build a responsive user interface that is clean with ReactJS that can facilitate smooth navigation and easy job exploration.
2. Re-implement a Django REST backend that can support safe authentication, resume upload, job advertisement and user profile administration.
3. To introduce skill-based job filtering that would enable users to find the relevant opportunities without unwarranted complexity.
4. To develop a resume and profile management system through which students can post, read, edit and manage their files comfortably.
5. To offer interview preparation materials, such as topic-related notes, job preparation documents, and preparation guides, curated.
6. To guarantee safe data processing with the help of the token-based authentication (JWT) and design-oriented privacy standards.

## 2.3 SCOPE OF THE PROJECT:

The CogniTrace is applicable in the whole process of the early-career recruitment cycle, that is to say the process of finding opportunities up to the preparation of interviews. It is meant to be beneficial to the students, fresh graduates, campus recruiters, training coordinators and academic institutions.

**For Students:**

- Skill-based job discovery
- Upload and management of resumes.
- The resources of interview preparation.
- Individual career guidance.

**For Academic Institutions:**

- An orderly platform to facilitate campus placement endeavors.
- Increased accessibility to student preparedness.
- Student skill profiles are centrally accessed.
- In the case of Training and Career Development Teams:
- Helps advise students according to job patterns and abilities.
- Serves as a preparation and learning aid.

The modular nature of the architecture of CogniTrace provides scope to expand in the future to include areas like:

- Internship portals
- Dashboards of the company-student communication.
- Computerized job suggestions.
- Government and industry job APIs integration.

In general, the system gives a solid framework to develop a scalable, student-centered recruitment platform that evolves in response to industry demands.

## 2.4 SIGNIFICANCE OF THE PROJECT:

The current-day students are not only asking about the list of vacancies they are also demanding transparency, orientation, and certainty in the dynamic and competitive job market. The traditional job portals overlook such requirements and early career navigation is a stressful and non-efficient process. CogniTrace will help to eliminate this problem with the help of a convenient to operate and student-centered platform.

**CogniTrace is also significant in:**

1. Empowering first time job seekers.
2. Through offering advanced job filtering, resume management and preparation tools.
3. Enhancing the employability preparedness.
4. With job discovery of skills and interview preparation materials.
5. Improving accessibility and transparency.

6. The students get a clear picture of the opportunities available and how those opportunities complement their skills.
7. Favoritism to data safety and security.
8. The system will make sure that personal information, resumes and profiles are treated with a great sense of responsibility.
9. Creating the distance between education and jobs.
10. CogniTrace assist students in moving the knowledge they have in the classroom to the career opportunity.

**As well, the platform supports the overall vision of learning institutions as it facilitates:**

- Better placement outcomes
- Profession-based career advice.
- Student support through technology.

By simplifying and streamlining the job search process and making it more accommodating, CogniTrace creates a future in which the students will feel prepared, knowledgeable, and job-ready; making recruitment not a stressful undertaking but a mentoring process.

# 3. RELATED WORK

## 3.1 INTRODUCTION:

In the current fast changing workforce, digital recruitment sites have served as the main connection between the employer and job seeker. Nonetheless, regardless of the presence of a significant number of job boards and hiring systems, students and recent graduates can hardly find the opportunities that would correspond to their skills and interests along with their career interests. The current platforms are recruiter-oriented and they apply a filtering method founded on key-word thus making it hard to get personalized and relevant job suggestions as well as support to early job hunters.

A CogniTrace is motivated by this major loophole in current recruitment systems. Graduates joining the job market need guidance, clarity and simplified job-searching experience, but most of them are bombarded on too many websites with generic listings, inappropriate suggestions, and limited information about their own positions. Although studies in the field of employability, digital credentials, artificial intelligence-based skill matching, and online hiring keep on expanding, existing systems are incapable of rendering these innovations to student-friendly systems.

In this chapter, the author brings out significant research works, technologies and recruitment trends that shaped up the creation of CogniTrace. It presents the shortcomings of conventional job platforms, the developments in AI-based job search and talent-based matching and why a talent-focused platform such as CogniTrace is crucial in contemporary digital hiring markets.

## 3.2 EXISTING SYSTEM AND LIMITATIONS:

Job listings and professional networking is available on a large scale through popular recruitment platforms like LinkedIn, Naukri, Indeed, and Glassdoor. These platforms are strong but are constructed majorly based on the needs of employers and not the needs of students or fresh graduates. Their recommendation systems are primarily based on profile keywords, search history, or job titles and very little attention is put on skill proficiency, academic background, or career stage.

The critical problems that students working with such platforms encounter include irrelevant job recommendations, absence of information about what is required of them, and lack of guidance that is personalized. These systems do not also incorporate digital credentials, achievement in projects or certification based skills which are very important when freshers join competitive job markets. Moreover, the majority of portals that are in place lack organized interview preparation, resume advice, and support at the start of a career.

**Key Disadvantages of the Current Recruitment Solutions.**

1. Matching using keywords alone - platforms cannot identify soft skills and certifications and changing student profiles.
2. Employer-centered design - algorithmic ranking focuses on the needs of the recruiter as opposed to the comprehensibility and experience of the candidate.
3. Minimal customization - the recommendations are not matched with the abilities, background of learning and the needs of the student.
4. No career guidance- systems do not provide interviewing tips, resume advice, and learning recommendations.
5. Poor assistance to freshers - fresh graduates are listed with wrong matches because of absence of experience related filtering.
6. Privacy issues - the use of personal data is not always clear, that is why students are reluctant to post resumes or profile information.

7. These loop holes indicate that a platform such as CogniTrace is required, which should be characterized by simplicity, transparency, and empowerment of students.

## 3.3 AI-BASED RESEARCH IN RECRUITING AND MATCHING SKILLS WITH JOB SKILLS AND EMPLOYABILITY:

Recent study e-recruitment, skill analytics and AI-based hiring has presented encouraging efficiency, equity and alignment of candidates and employment. Other studies like Faliagka et al. (2012) proposed automated applicant ranking and personality mining, whereas Ghazarian et al. (2021) conducted a review of machine-learning methods of job recommendation and skill-matching. Those works prove that data-driven hiring may significantly increase the performance of job suggestions, but it works only when the profile of the candidates is perceived as a whole.

The shift towards the significance of soft skills and alternative learning methods is supported in research studies on graduate employability (Andrews and Higson, 2008). MOOCs and micro-credentials (Chakraborty et al., 2020) are becoming useful indicators of job-readiness, yet big job portals do not implement them well.

The articles by Jeske and Shultz (2016) write about the emergence of social media screening in the recruitment process and mention the issues of privacy, fairness, and bias in particular, which are particularly important to consider when related to students, as their online behavior may be misunderstood. In the meantime, Nawaz et al. (2022) emphasize the benefits of recruiting tools based on AI to address skills deficits, yet warns about opaque algorithms that hurt freshers.

Lastly, the Future of Jobs Report (2020) by the World Economic Forum highlights that future employment activities will rely on new abilities, digital skills, and lifelong learning, which are the primary trends in job recruitment, whereas current portals are not updated with these trends.

The overall outcome of these studies is that the technology of recruitment is progressing, but the elements of student-centered customization, skill-based match and ethical data usage remain missing, and they become the basis of the approach of CogniTrace.

## 3.4 COMPARATIVE ANALYSIS OF EXISTING APPROACHES:

| Platform | Approach | Strengths | Limitations |
|---|---|---|---|
| **LinkedIn** | Professional networking & algorithmic job suggestions | Industry insights, recruiter outreach | Overwhelming for freshers, limited student-centric filtering |
| **Naukri** | Keyword-based job search | Huge database, filters | Too generic, lacks personalized recommendations |
| **Indeed** | Aggregated job listings | Easy to use, global listings | Minimal skill-based matching |
| **Campus Portals** | Recruiter-controlled job postings | Useful for placement drives | No personalization or continuous learning path |
| **CogniTrace (Proposed)** | Skill-based filtering + student-first job recommendations + privacy-focused design | Personalized, simple, transparent, supportive for freshers | Large-scale AI-based recommendation planned for future |

## 3.5 IDENTIFICATION OF RESEARCH GAP:

Based on the analysis of the current platforms and research papers, the following gaps can be identified:

1. Inability to be skill-based when it comes to individualizing the students and new graduates.
2. Relies on the use of keywords more than a deep skill/job match.
3. Lack of attention to the digital certificates, MOOCs, academic projects, and domain interests.
4. Lack of legal advice like resume tips or interview preparation.
5. Lack of transparency regarding the use of data and privacy, particularly among freshers.
6. The design of recruiter-first, which causes anxiety, confusion, and overload of information among students.

**The way CogniTrace Handles These Gaps:**

1. Provides a student-first culture that emphasizes transparency and assurance.
2. It offers skill-based filtering in accordance with user preferences and abilities.
3. Protects privacy-aware profile operation at transparent use of data.
4. Combines uploading of resumes, interview preparation and easy navigation to assist in early-career requirements.
5. Prepares the future AI-based job suggestions and personalised learning recommendations.

# 4. PROBLEM STATEMENT

## 4.1 INTRODUCTION:

In this digital and competitive world of employment, more so in the job seeking market, students and freshly graduated job seekers are overly relying on the internet to search the job market. Whereas job portals including LinkedIn, Naukri and Indeed offer access to thousands of listings, they are more of generalized recruiter systems. The emphasis placed on these platforms is on the needs of employers and matching based on keywords, and does not provide much information of the skill set of the applicants, their academic progress and the level of readiness.

New graduates usually face difficulties in finding same job opportunities that match their interests, skills and career objectives. They face several obstacles including inappropriate job advice, lack of customized advice, job expectations, and scarce resources on how to build resumes or prepare to conduct an interview. Consequently, the job-search process turns out to be daunting, baffling, and time-consuming.

CogniTrace is created to handle such issues as it provides a student-focused recruitment platform with emphasis on clarity, personalization, and accessibility. In contrast to the usage of keywords as the only type of filters, CogniTrace focuses on skill-based matching, guided search, resume management tools, and systematized interview preparation tools. The platform will make the job-hunting process simpler with the help of structured job information and user-friendly workflows that allow students to make knowledgeable career choices without doubt.

## 4.2 STATEMENT OF THE PROBLEM:

The conventional web based recruitment sites are not designed with students in mind-only employers. Their emphasis lies on mass posting of job advertisements, screening of resumes and filtering applications, but they do not offer much assistance or customization to fresh graduates who are only entering the job market. Most students get lost in the process and are not able to find the right opportunities and job expectations.

**The current digital recruiting process does not solve three essential needs of people trying to find a job in an early career:**

1. **Absence of Competency-based Individualization:** Majority of platforms work through generic matches based on keyword or title regardless of the real skills and certifications, projects and learning background of the student.

2. **Absence of Student-Centred Advice:** Resume building, skill development and interview preparation- aspects that fresh graduates desperately require are not structured in any way.

3. **Absence of Transparency and Simplicity:** It is usually a stressful and non-intuitive process where students often get irrelevant suggestions and cannot tell how the workflow is followed in the applications.

**Thus, the essence of the problem is as follows:**

To create and develop a recruitment platform that offers skill-based, transparent, and student-centered job-search experience, allowing fresh graduates to search and evaluate the opportunities, handle profiles, and prepare to placements with the help of an efficient and supportive online experience.

This issue highlights the increased necessity of recruitment systems that extend beyond simple job posting and are proactive towards empowering the early career job seekers.

## 4.3 NEED FOR THE STUDY:

The process of making a transition between academic and professional life is among the most crucial stages of the student life. The majority of the recruitment platforms that are currently available do not cater to people at this early phase, however. Students often experience:

**Problem locating the right entry-level jobs.**

- Poor knowledge of skills needed and job requirements.
- Poor resume, portfolio and interview preparation resources.
- Deficit of confidence caused by bad guidance and being over-choosing.
- This study is necessitated by the gap between the requirements of the students and the offerings of most job portals today.

**CogniTrace will fill this requirement by:**

- Emphasis on job matching based on capability.
- Providing resume posting and profile services.
- Presenting documents on interview preparation.
- Providing a plain, uncluttered and user-friendly interface.
- Having privacy and transparency in the process of user information management.

The objective of the platform is to minimize the confusion, enhance the relevance of the job and also direct the students in the hiring process- developing a systematic direction between the profile creation and submission of the application.

## 4.4 OBJECTIVES OF THE STUDY:

The primary mission of CogniTrace is to create a specific student/fresh graduate recruitment website so that it is easy to navigate, search and apply in terms of skills required, find jobs and make applications easy.

**The specific objectives are:**

1. To build a skill-based job recommendation system, which will assign users to jobs based on their skills, interests, educational background as per their skills and interests.
2. To create a safe Django-based backend to facilitate authentication, profiles, resume submissions and job advertisements.
3. To create a responsive React-based interface that provides a smooth navigation experience, the well-arranged dashboards and the user-friendly experience.
4. To offer interview preparation contents such as papers, tools and topic-based study contents to aid early career preparedness.
5. To facilitate resume management where the user can upload, update and download resumes safely.
6. To guarantee privacy settings that are student-centric and open management of user data.
7. All these goals are meant to help the job seekers in the early career stages to have the tools and understanding on how to go through the hiring proceFss effectively.

## 4.5 SCOPE OF THE STUDY:

CogniTrace has a broad range of the end-to-end job-search experience of the student, including profile development and skill-based job-search, resume submission, and interview preparation.

**Students & Fresh Graduates:** Users can create profiles upload resumes see what is available based on abilities, position, and location. Distinct preparation materials of an interview. Be prepared when it comes to job applications.

**Future Scope:** Training & Placement Cells / Institutions.

**The system may be expanded so that it will enable the colleges to:**

- monitor student skills
- track application activity
- manage placement drives

**In the present research, the prototype is aimed at developing a complete operating prototype that has:**

- Skill-based filtering
- Resume uploads
- Account management
- Job listing module
- Interview preparation part.
- Secure authentication

**The architecture however enables extension in the future which includes:**

- AI-powered recommendations
- External job API integration.
- Analytics dashboards
- Mobile app development

CogniTrace is a practical solution that can be used at this point in time and a solid base on the advanced recruitment technologies in the future, therefore.

# 5. REQUIREMENT ANALYSIS, RISK ANALYSIS, FEASIBILITY ANALYSIS

The design for a recruitment system like CogniTrace needs to analyze the needs of the system, potential risks of the system, and determine whether the project is viable both technically and operationally. CogniTrace is an online job search and recruitment platform targeting students, with the purpose of making the job search process easier among fresh graduates

**This chapter outlines:**

- Non-functional requirements and functional requirements.
- System specifications
- The associated risks and mitigation strategies.
- It must be economically, technically, operationally, and legally viable.

This discussion treats CogniTrace as viable, valid, scalable, and capable of addressing reality with regard to the problems students face concerning employability in the real world.

## 5.1 REQUIREMENT ANALYSIS:

In the requirement analysis, the functionalities to be performed by the system-that is, functional requirements-and the standards to be met by the system-that is, non-functional requirements-are identified. The CogniTrace is meant to offer skills-based job-filtering, resume management, secure authentication, and a user-friendly interface.

### 5.1.1 Functional Requirements:

These requirements specify the activities that CogniTrace must perform.

1. User Authentication
   - The system should be able to register and log in to email securely, with verification of OTP.
   - JWT-based authentication provides access that is secure and in a session.
2. Resume and User Profile Management.
   - Users should be in a position to create/update profiles.
   - Resumes shall be uploaded/downloaded in the system in PDF/Docx formats only.
   - The information in the resume should be kept in a safe place where it can be accessed anytime.
3. Competency Management
   - It allows users to add, modify, remove, and confirm skills.
   - Possibility to attach skills certificates (PDF/JPG).
4. Job Postings and Skill-based Filtering.
   - It must be able to fetch and display the job postings whenever needed.
   - The users should be given job recommendations based on their skills, location, and category of job. There should be a search and filter facility.

5. Interview Preparation Module.

   - The system will comprise categorized interview preparation materials such as Java, Python, Cloud, DBMS, Data Structures. A facility for the user to access/download study materials should be provided.

6. Assessment Module

   - CogniTrace is supposed to redirect to other certification sites: Hackerrank, NPTEL, etc.
   - Log evaluation efforts or stored certificates (not mandatory).

7. Notifications and Settings

   - The platform shall support new job alerts, profile update notifications, and assessment reminders.
   - It allows users to customize and set visibility options accordingly.

8. Help & Support

   - A FAQ section and support email should be provided in the system to handle queries from users.

### 5.1.2 Non-Functional requirements

The requirements specify the quality, performance and usability standards of the system.

1. Security

   - Encryption for all sensitive data should be provided.
   - JWT authentication needs to be implemented.
   - Safe management of resumes and certificates.

**2.** Usability

- The interface of the application must be easy to use, intuitive, and reactive.
- Smooth navigation should be provided between the dashboard, jobs, resume, and skills.

**3.** Performance

- The job postings and the profile information should load within 1 to 2 seconds.
- The platform must support many users operating the platform simultaneously.

**4.** Scalability

- Future expansion supported by the Django REST backend shall include handling large databases, integrating additional job APIs, and enabling AI-based recommendations.

**5.** Reliability

- It needs to have a 99 percent availability.
- Uploading/downloading of the resumes should be reliable, and data should not get lost.

**6.** Compatibility

- CogniTrace is likely to run on Chrome, Edge, and Firefox browsers across Android/iOS devices and laptops/desktops.

### 5.1.3 Personas  Requirements

**Persona 1: Student User**

- **Age:** 22
- **Goal:** Quick and simple interaction
- **Needs:** Easy navigation with minimal steps
- **Frustration:** Confusing layouts

**Persona 2: Working Professional**

- **Age:** 35
- **Goal:** Fast task completion
- **Needs:** Efficient and responsive system
- **Frustration:** Slow performance

**Persona 3: Beginner User**

- **Age:** 18
- **Goal:** Learn and use the system easily
- **Needs:** Simple instructions and clear interface
- **Frustration:** Difficulty understanding complex features

**Persona 4: Elderly User**

- **Age:** 55+
- **Goal:** Use the system without confusion
- **Needs:** Large buttons, readable text, simple design
- **Frustration:** Small text and complicated navigation

**Persona 5: Frequent User**

- **Age:** 28
- **Goal:** Perform tasks quickly on a regular basis
- **Needs:** Shortcuts and fast interaction
- **Frustration:** Repetitive steps

**Persona 6: Accessibility-Focused User**

- **Age:** 30
- **Goal:** Use the system comfortably with minimal strain
- **Needs:** Clear visuals, good contrast, easy interaction
- **Frustration:** Poor accessibility design

### 5.1.4 SYSTEM REQUIREMENTS:

| Category | Specification |
|---|---|
| **Frontend** | ReactJS, JavaScript, Tailwind CSS |
| **Backend** | Django, Django REST Framework |
| **Database** | SQLite3 (development), PostgreSQL (optional future upgrade) |
| **Authentication** | JWT |
| **Tools** | VS Code, Git, GitHub |
| **Hosting** | Render / Vercel / Railway (future deployment) |

## 5.2 RISK ANALYSIS:

| Risk Type | Description | Probability | Impact | Mitigation |
|---|---|---|---|---|
| **Technical Risk** | Job APIs may fail or return slow responses | Medium | Medium | Implement caching & fallback data |
| **Data Privacy Risk** | Resume and personal info leakage | Low | High | Encrypt data, secure JWT, HTTPS |
| **Performance Risk** | Delay in loading jobs for many users | Medium | Medium | Optimize queries & server load |
| **User Adoption Risk** | Freshers may find UI confusing | Medium | Medium | Conduct UI testing & simplify flows |
| **Operational Risk** | Server downtime or deployment issues | Low | High | Cloud backup, monitoring |
| **Integration Risk** | Errors in linking assessments or external job portals | Medium | Low | API versioning & testing |
| **Legal Risk** | Misuse of user data | Low | High | Transparent privacy policies |

## 5.3 FEASIBILITY ANALYSIS

This feasibility study will investigate the viability, practicality, and sustainability of deploying CogniTrace as a full-scale recruitment and career-support platform to students and early professionals. It also appraises the possibilities of designing, developing, deploying, and maintaining the proposed system using the available resources, technologies, and institutional support.

The paper has examined four major dimensions such as technical feasibility, economic feasibility, operational feasibility and legal and ethical feasibility in order to determine whether CogniTrace could fulfill its purpose given realistical grounds. Each of the dimensions is essential in understanding the strengths of the system, the potential limitations and estimating the system lifespan in the actual real world academic and recruitment situations.

The reason behind this feasibility analysis is to make sure that CogniTrace is theoretically right but at the same time practically, cost-effective and useful to job seekers and educational institutions. It is with this multi-layered evaluation that it is confirmed that CogniTrace could be formulated as a scaling, secure and high-impact platform that can help students in a significant way to the career-building process.

### 5.3.1 TECHNICAL FEASIBILITY

1. CogniTrace can be technically implemented because:
    - It applies highly adopted technologies: React, Django, SQLite.
    - React goes hand in hand with Django REST API.
    - It's easy to deploy with Render or Vercel.
    - Resume upload and job filtering are lightweight operations.

2. The modular architecture permits:
    - Adding new job APIs
    - AI-powered job recommendation modules.
    - Mobile Application Expansion

### 5.3.2 ECONOMIC FEASIBILITY:

CogniTrace uses open-source tools → **low development cost**.

| **Cost Component** | **Estimated Cost (INR)** |
|---|---|
| **Development Tools** | Free (Open Source) |
| **Hosting (Render/Vercel)** | ₹5,000 – ₹12,000/year |
| **Domain Purchase** | ₹800 – ₹1,200/year |
| **Maintenance** | Minimal |

### 5.3.3 OPERATIONAL FEASIBILITY:

1. CogniTrace is easy to use because:
    - Students can work their way in home, resume, skills, and jobs sections.
    - Jobs can be easily administered by the recruiters themselves, feature of the future.
    - Simple UI involves less training.
2. Cloud deployment ensures High availability, Fast updates and Easy scaling

### 5.3.4 LEGALITY AND ETHICAL VIABILITY:

1. CogniTrace ensures Safe data storage of resumes and users.
2. No personal information was mishandled. Open data management practices Unbiased ethical filtering. Compliance in the future may involve: Privacy statements compliant with GDPR. AI fairness policies

# 6. PROPOSED SOLUTION

## 6.1 INTRODUCTION:

Most traditional recruitment portals are designed to be recruiter-centric; hence, they focus on bulk hiring, filtration through keywords, and employer-driven processes. Such portals do little to understand the needs of students and freshers entering the job market for the first time. These beginners get irrelevant jobs listed for them, find the filters confusing, and get no personalized support.

CogniTrace seeks to address this challenge through the provision of a simple, student-first, skill-based job-search platform focused on clarity, personalization, and usability. On the site, users can create profiles, upload resumes, manage skills, and browse job opportunities that truly fit their education, preferences, and competencies.

Designed on the basis of ReactJS frontend and Django REST Framework backend, CogniTrace guarantees the clean interface, high-performance, and security of user information processing. It focuses on user-centered design, which allows recent graduates to have custom job filters, clear navigation, and open data application. The vision is to provide a platform where students have confidence and control of their career path, which is aware of them, their needs, interests, and their skillsets.

## 6.2 OVERALL SYSTEM ARCHITECTURE:

It has a modular and scalable architecture design to enable secure authentication, skill management, resume uploading, and filtering of jobs based on skills. Further, the CogniTrace architecture includes the following layer system:

1. Frontend Interface - Client Layer
    - Implemented using ReactJS
    - Provides UI for resume upload, skills, job listings, and interview preparation
    - Communicates with backend via REST API calls
    - Implements role-based navigation and protected routes
2. Application Logic Layer
    - Manages the user flows: registration → resume → skills → jobs
    - Handles filtering of jobs and navigation between pages.
    - Manages state using React hooks / Context API
3. Backend & API Layer: Django REST Framework
    - Provides REST APIs for authentication, resume upload and job retrieval.
    - Validates user inputs
    - Manages CRUD for skills and user data
    - Stores uploaded resumes and profile details.
4. Database Layer (SQLite3)
    - Maintains relations between users → skills → resumes

5. Data Processing & Filtering Layer
    - Handles skill-based filtering
    - Ensures early-career-friendly recommendations
6. Optional Future AI Layer
    - Planned for future enhancements
    - AI-based job matching using ML
    - Career recommendations
    - Resume suggestions

## 6.3 PROPOSED METHODOLOGY:

The workflow of CogniTrace allows users to register smoothly, upload their resumes, manage their skills, and look for suitable job listings.

**Steps:**

1. User Registration & Authentication
    - JWT-based login & signup
    - Email stored locally for personalization
2. Creating Profiles
    - Users input basic information
    - Upload resume file (PDF/DOCX)
    - Skills can be added dynamically

**3.** Competency Management

- Add / update / delete skills
- Backend stores skills linked to user ID
- Skills are used for filtering jobs recommended to a candidate.
- API returns jobs
- The backend filters results based on Skills, Job type and Location (if added later).
- Technology preferences

**4.** Displaying Jobs

- Clean grid/card layout
- Call-to-action buttons: Apply, Save, View Details
- Links to external job applications if necessary

**5.** Interview Preparation Module

- Provides resources for Java, Python, Cloud, DSA, Aptitude etc..
- Cards with downloadable notes
- Resumes stored in the Django media folder
- User data and skills are stored in SQLite3.
- Only authenticated users can see their own data

## 6.4 UNIQUE FEATURES OF COGNITRACE:

1. Student-First Design
2. Tailor-made for freshers, not recruiters.
3. Skill-Based Job Filtering
4. Provides relevant and focused opportunities.
5. Smooth Resume Upload System
6. Secure file handling and easy storage.
7. Modern UI/UX
8. Clean, simple, distraction-free interface.
9. Interview Preparation Module
10. Privacy-Focused Implementation
11. Data use that is transparent and authentication that is secure.
12. AI-Ready Architecture
13. Future support for job recommendation models.

## 6.5 DESIGN PRINCIPLES FOLLOWED:

1. Easy navigation
    - Minimal steps
    - Beginner-friendly terminology
2. Modular System Design
    - Separate skills module from the Job module
    - Authentication independent of job filtering
    - Easier to maintain and extend
3. Scalability
    - Supports future AI job matching
    - Can be migrated to PostgreSQL or cloud.
    - React components reusable and maintainable

## 6.6 TECHNOLOGY STACK:

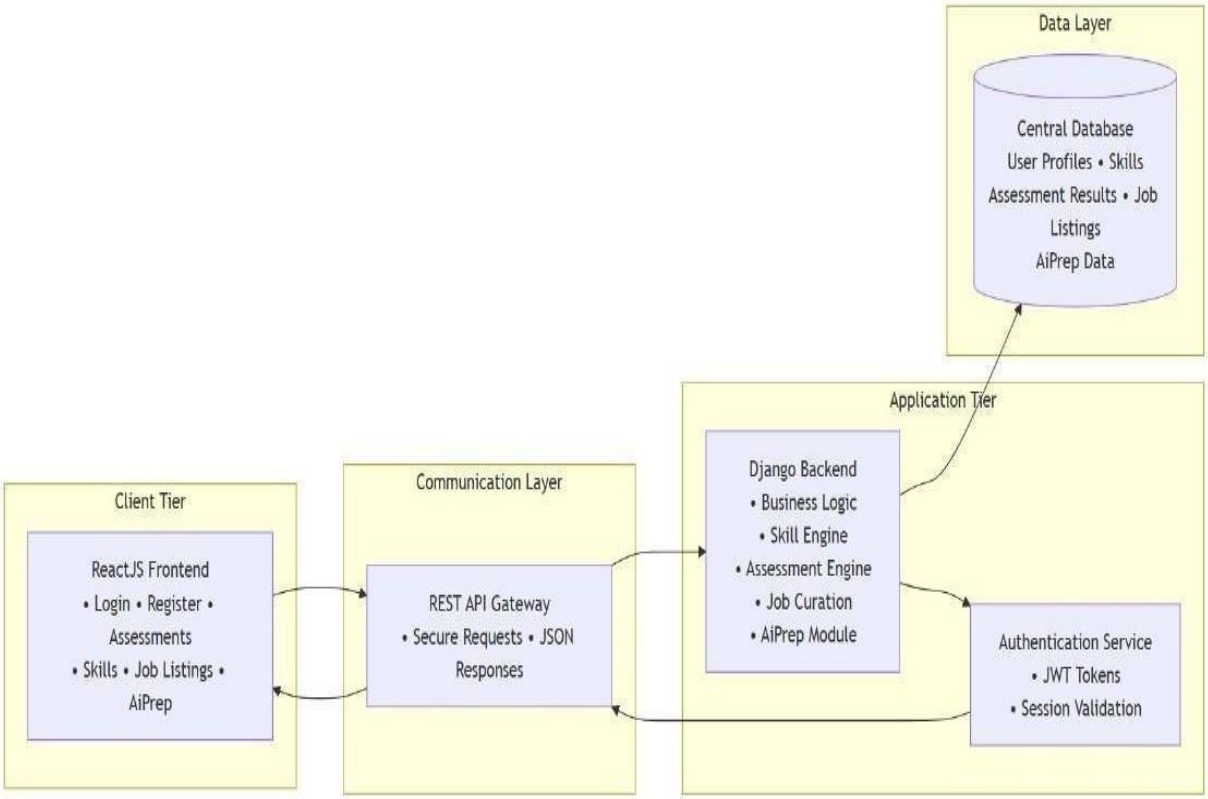
| Component | Technologies Used | Purpose |
|---|---|---|
| **Frontend** | ReactJS, CSS, React Router | UI for job search & resume upload |
| **Backend** | Django, DRF | APIs, authentication, data management |
| **Database** | SQLite3 | User + job + resume storage |
| **Authentication** | JWT | Secure login |
| **Storage** | Django Media | Resume & file uploads |
| **Version Control** | GitHub | Code hosting |
| **Deployment** | (Optional) Render, Vercel | Production hosting |

## 6.7 SYSTEM INNOVATION:

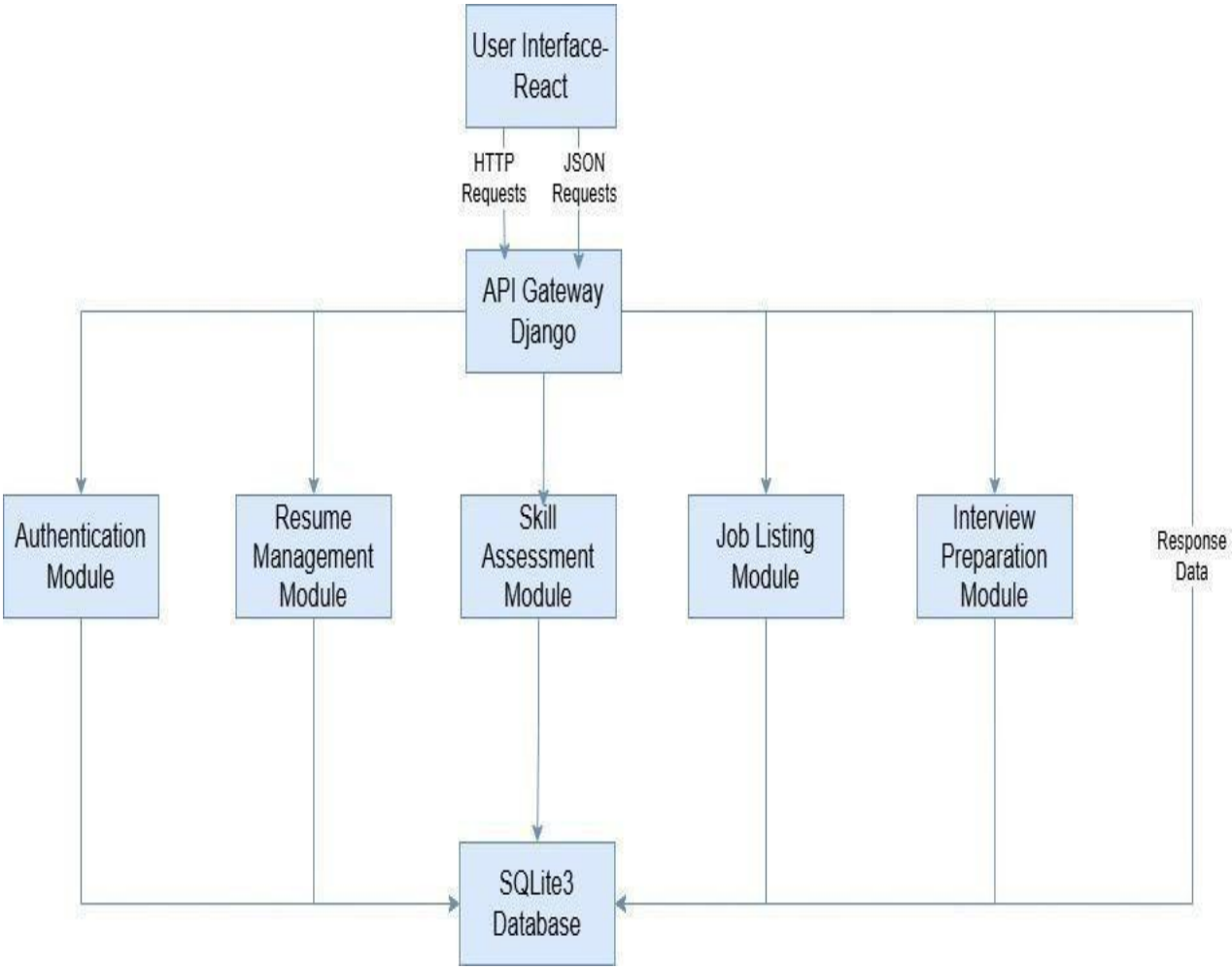
CogniTrace introduces a student-first recruitment model. Unlike traditional portals, it: Helps freshers to discover jobs that match their skills. Reduces overwhelm by clean filtering Encourages career development through structured interview preparation Ensures transparency and data safety Provides a platform that grows with the user. Over time, CogniTrace can integrate AI models to: Predict best job matches Recommend certificates / courses Skill improvements that may be suggested

# 7. ARCHITECTURE DIAGRAM, FLOW CHART

## 7.1 ARCHITECTURE DIAGRAM:



## 7.2 FLOW CHART:



# 8. SIMULATION SET UP AND IMPLEMENTATION

**Github Link: https://github.com/Evancy2460/2503B05116_HCID_CogniTrace-Job-Search-Recruitment-Platform-Project.git**

## 8.1 SOFTWARE & HARDWARE CONFIGURATION:

| Component | Specification |
|---|---|
| **Processor** | Intel Core i3 / i5 / i7 |
| **RAM** | 8 GB minimum |
| **Storage** | 256 GB – 512 GB SSD |
| **Operating System** | Windows 10 / 11 |
| **Frontend Framework** | React.js, JavaScript (ES6+), Tailwind CSS |
| **Backend Framework** | Django, Django REST Framework |
| **Database** | SQLite3 |
| **Server Runtime** | Python 3.10+ |
| **APIs & Libraries** | REST Framework, JSON Web Tokens (JWT), Django ORM |
| **Frontend Tools** | Axios, React Router, React Icons |
| **Version Control** | Git & GitHub |
| **Development IDE** | Visual Studio Code |
| **Testing Tools** | Postman / Thunder Client |
| **Deployment Platform (Optional)** | Render / Railway / Vercel (Frontend) |

**Tech Stack Used**

- Backend: Django (Python) with Django REST Framework for API development.
- Frontend: React.js for building the user interface.
- Database: SQLite3 (default for Django projects, suitable for development and testing).
- Authentication: JWT (JSON Web Tokens) for secure token-based authentication.

## 8.2 FRONTEND COMPONENTS:

**1. Skills Component**

**Function:** Manage user skills for profile building and job matching.

**Features:**

- Display the current list of skills.
- Add, edit, and delete skills.
- Use validations to ensure correct skill format.

**Implementation Details:**

- On component mount, fetch skills from backend API.
- Provide options to update/delete skills via PUT/DELETE requests.
- Use React hooks to manage skill list state, loading, and error feedback.

**2. Assessments Component**

**Function:**

Facilitate users to access and take HackerRank certification tests directly through the CogniTrace

platform.

**Features:**

- Display list of available HackerRank certification tests or links.
- Provide access/redirection to HackerRank's platform for test taking.
- Optionally show test status, scores, or certificates fetched from HackerRank APIs after completion.
- No internal test question rendering or submission is implemented.

**Implementation Details:**

- Fetch the curated list of HackerRank certification tests or their URLs from your backend API or configuration (GET /api/assessments/ may return test metadata with external links).
- Use React components to render test names and provide direct links/buttons that open HackerRank test pages in new tabs/windows.
- No internal logic for question navigation, answer collection, or scoring submission since tests are fully hosted and managed by HackerRank externally.

**3. Interview Prep Component**

**Function:** Provide users with interview preparation materials (documents and guides).

**Features:**

- Display curated and related documents for interview preparation.
- Organized by topic or skill area.
- Simple document viewer or downloadable links.

**Implementation Details:**

- Fetch interview prep documents and related resources using API.
- Render documents as lists or cards with links for viewing or downloading.
- No interactive practice or quizzes; purely informational content.

**4. Home Component (Skills-Based Job Listings)**

**Function:** Display curated job listings filtered by user skills.

**Features:**

- Fetch and display jobs matching user skills.
- Search and filter jobs by criteria such as location or experience.
- Job card UI with summary and apply/save buttons.

**Implementation Details:**

- Call backend API (GET /api/jobs/skillbased/) to retrieve curated jobs.
- Render jobs dynamically using reusable job card components.
- Support user interactions for applying or saving jobs.
- Refresh or pagination controls to manage data volume.

**5. Help Component**

**Function:** Provide user support via a help email contact.

**Features:**

- List frequently asked questions (FAQs).
- Show a simple UI with the help email address for user queries.
- No ticketing or chat systems included.

**Implementation Details:**

- Fetch FAQ data from backend (GET /api/help/faqs/) for display.
- Show static help email address visibly for user support contact.
- Provide instructions or guidelines for users to send help mails externally.
- No backend form submission or ticket creation.

## 8.3 DATABASE DETAILS:

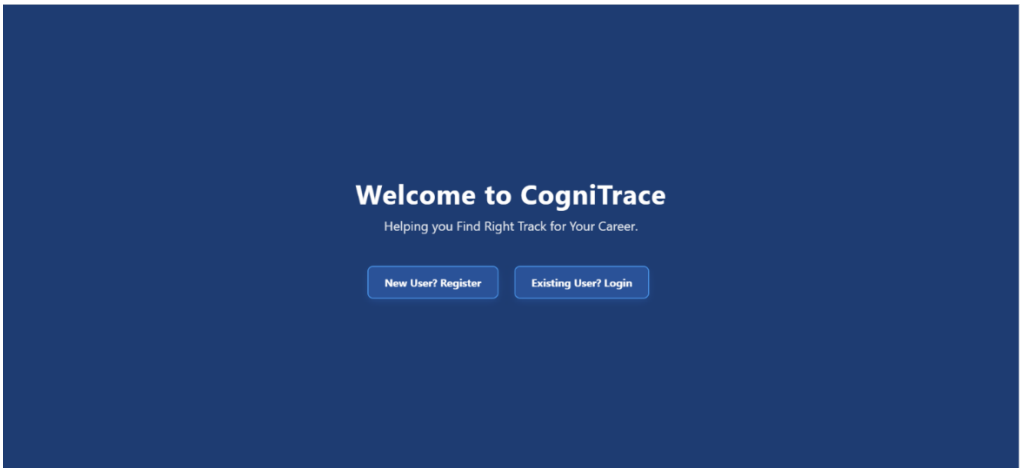
**Users:** The Users table holds all the essential information about each person using CogniTrace. It includes basic details like their username, email, and password-securely stored. In addition to that, it contains all their profile details including their name and contact details. Among the significant characteristics of this table, the fact that it handles files like resumes and skill certificates uploaded by the users should be mentioned. All these files are attached directly to the account of every user so that the important documents could be followed easily. The system logs the time such files were uploaded and retains the initial names of the files so as to keep things orderly and easily accessible.

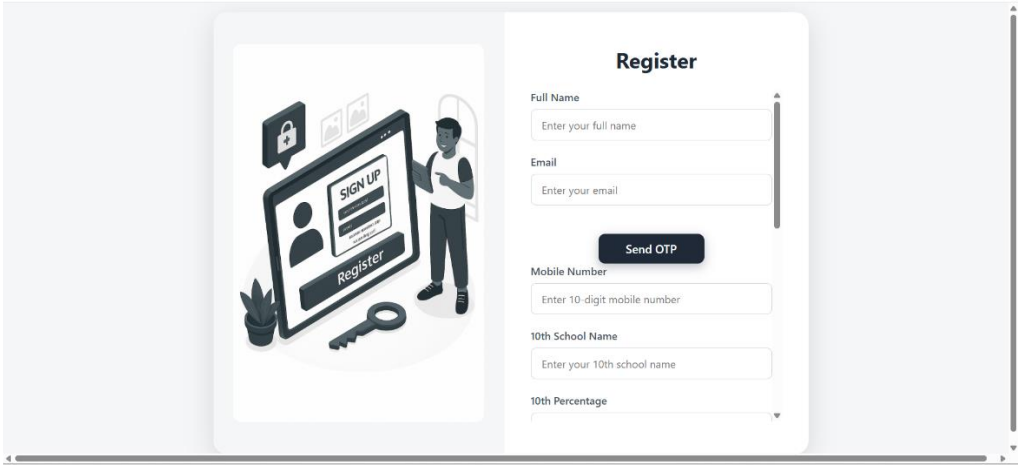
**Skills**: every skill added by a user is saved in Skills table. With every skill, we store data such as the name of the skill, the degree of proficiency that the user has with reference to that particular skill as well as whether the skill is verified or not. Authentication is achieved through the attachment of certificates, which are also stored as files, which are attached to the specific skill or directly to the profile of the user. This way, CogniTrace can with confidence reveal what skills users really possess, supported with the right documentation if necessary. Saving and managing such certificate files within the system is integrated, hence making the upload processes smooth and secure.

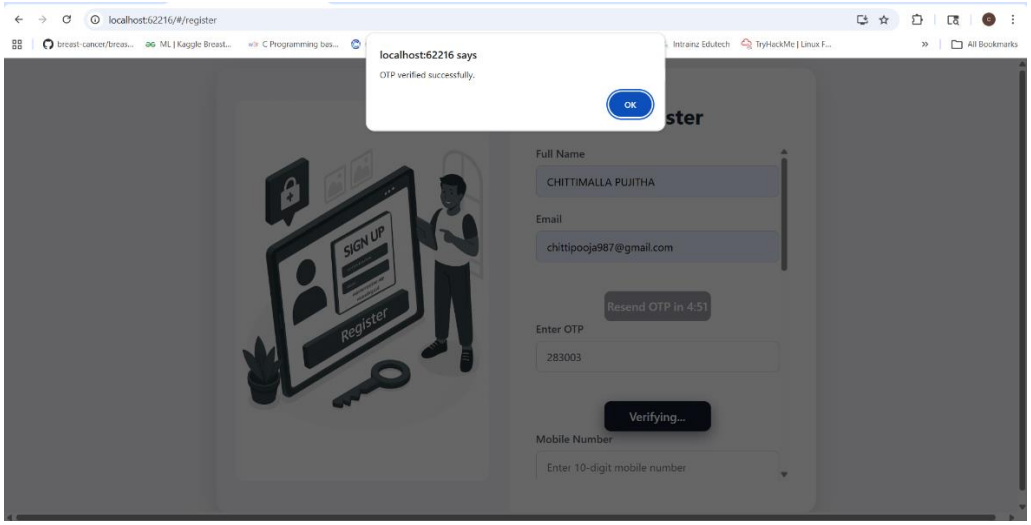
## 8.4 BACKEND COMPONENTS:

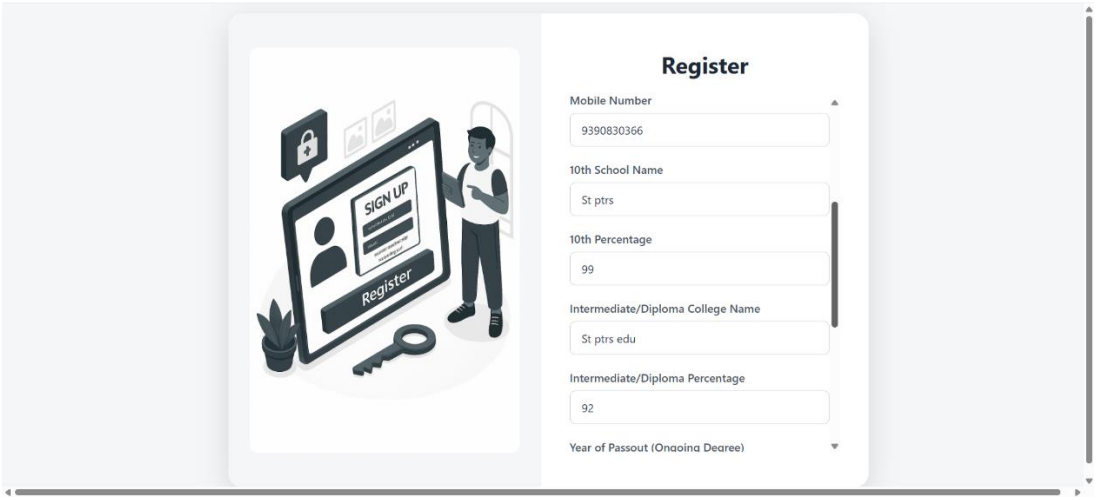
| S. No. | Functionality | Endpoint | HTTP Method | Description |
|---|---|---|---|---|
| 1 | Send OTP (Registration) | /send-otp-registration/ | POST | Send OTP for new user registration |
| 2 | Send OTP (Login) | /send-otp-login/ | POST | Send OTP for user login |
| 3 | Verify OTP | /verify-otp/ | POST | Verify the OTP submitted by user |
| 4 | Register User | /register/ | POST | Register a new user |
| 5 | Reset Password | /reset-password/ | POST | Reset password |
| 6 | Login | /login/ | POST | User login |
| 7 | Get User Details | /get-user/ | GET | Retrieve authenticated user details |
| 8 | Get Skills | /skills/ | GET | Get the list of user skills |
| 9 | Add Skill | /skills/add/ | POST | Add a new skill |
| 10 | Delete Skill | /skills/delete/<int:skill_id>/ | DELETE | Delete a skill by skill ID |
| 11 | Verify Skill | /skills/verify/<int:skill_id>/ | POST | Verify a specific skill |
| 12 | Upload Skill Certificate | /skills/upload/<int:skill_id>/ | POST | Upload skill certificate file |
| 13 | Upload Resume | /resume/upload/ | POST | Upload user resume file |
| 14 | Download Resume | /resume/download/ | GET | Download user resume file |
| 15 | Delete Resume | /resume/delete/ | DELETE | Delete user resume |
| 16 | Get Resume | /resume/get/ | GET | Get user resume file or metadata |

# 9. RESULT ANALYSIS




Register
Mobile Number
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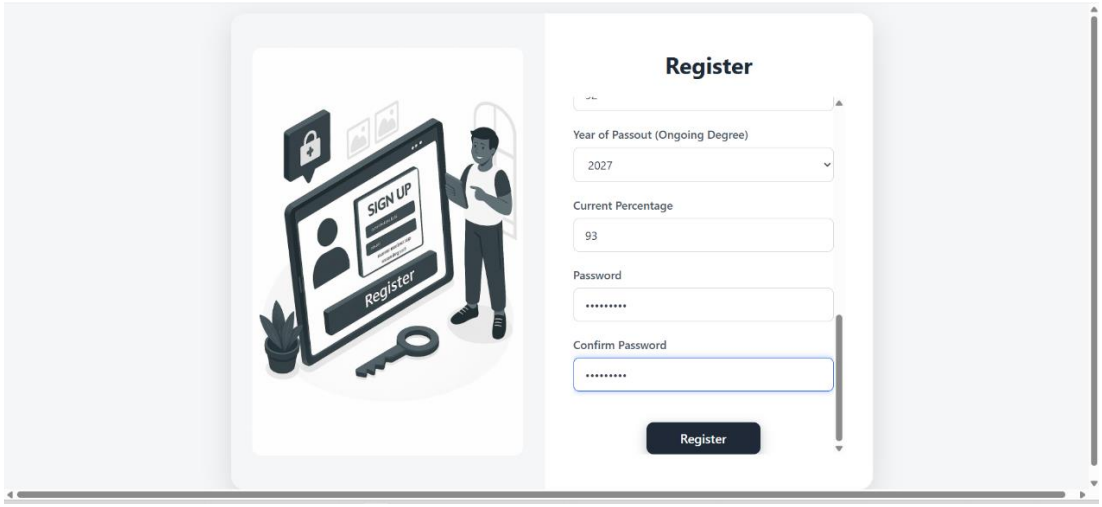
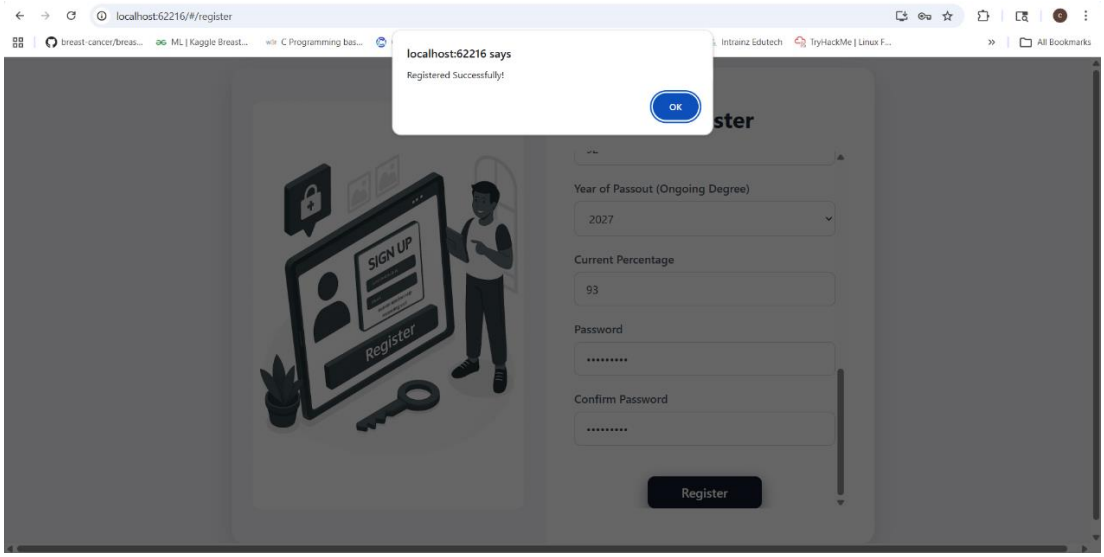
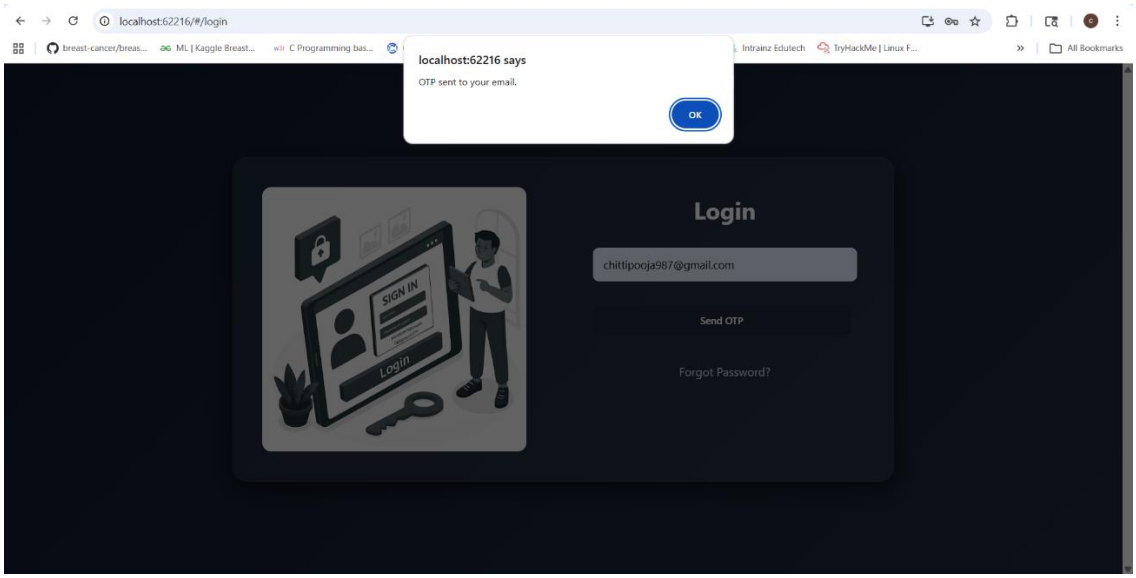
10th School Name
St ptrs
10th Percentage
99
Intermediate/Diploma College Name
St ptrs edu
Intermediate/Diploma Percentage
92
Year of Passout (Ongoing Degree)
SIGN UP
Register


Register
Year of Passout (Ongoing Degree)
2027
Current Percentage
93
Password
Confirm Password
Register
SIGN UP
Register


localhost:62216/#/register
localhost:62216 says
Registered Successfully!
OK
Year of Passout (Ongoing Degree)
2027
Current Percentage
93
Password
Confirm Password
Register
SIGN UP
Register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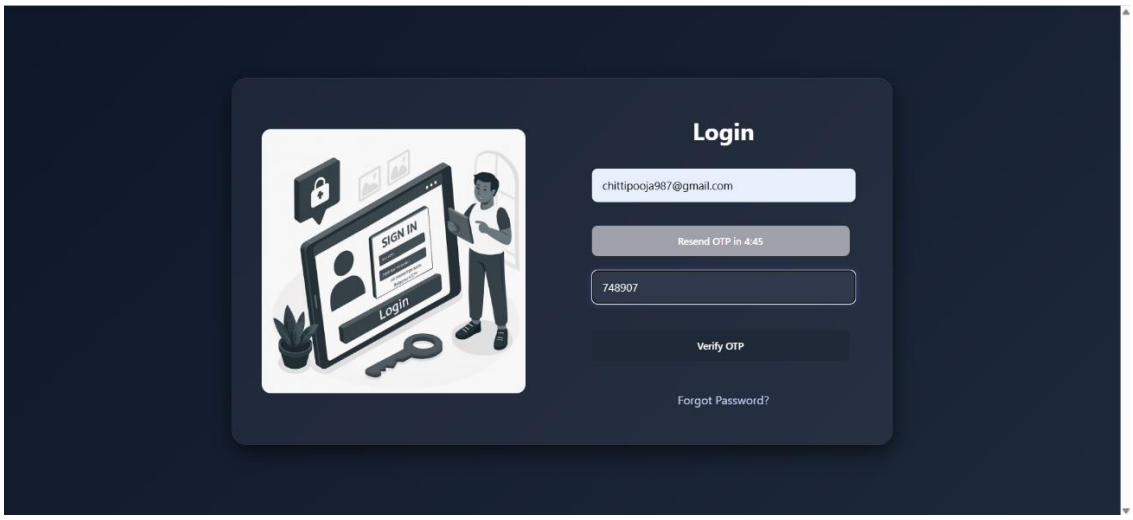
OTP sent to your email.
OK
Login
chittipooja987@gmail.com
Send OTP
Forgot Password?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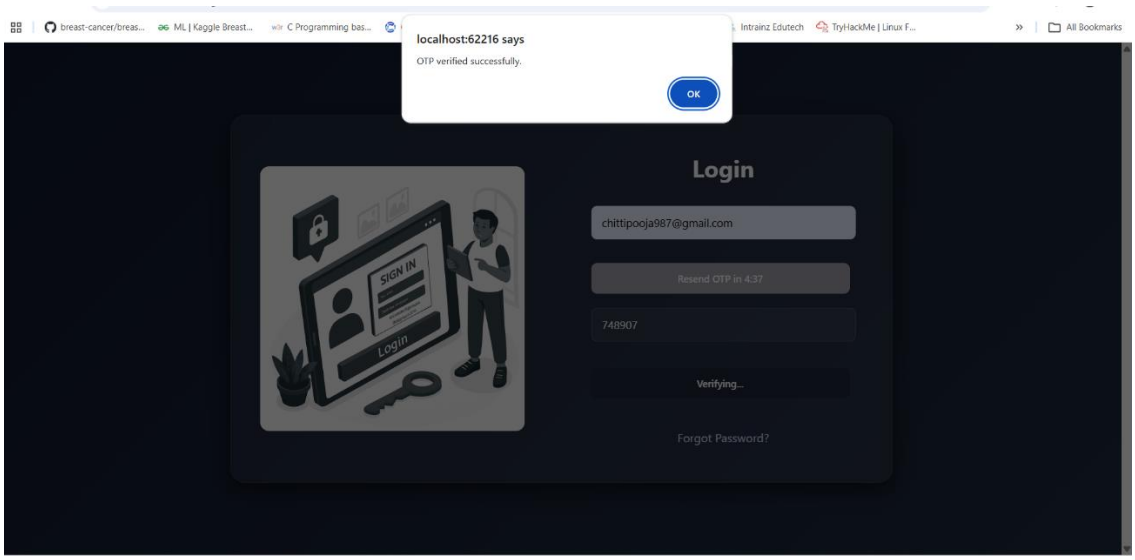
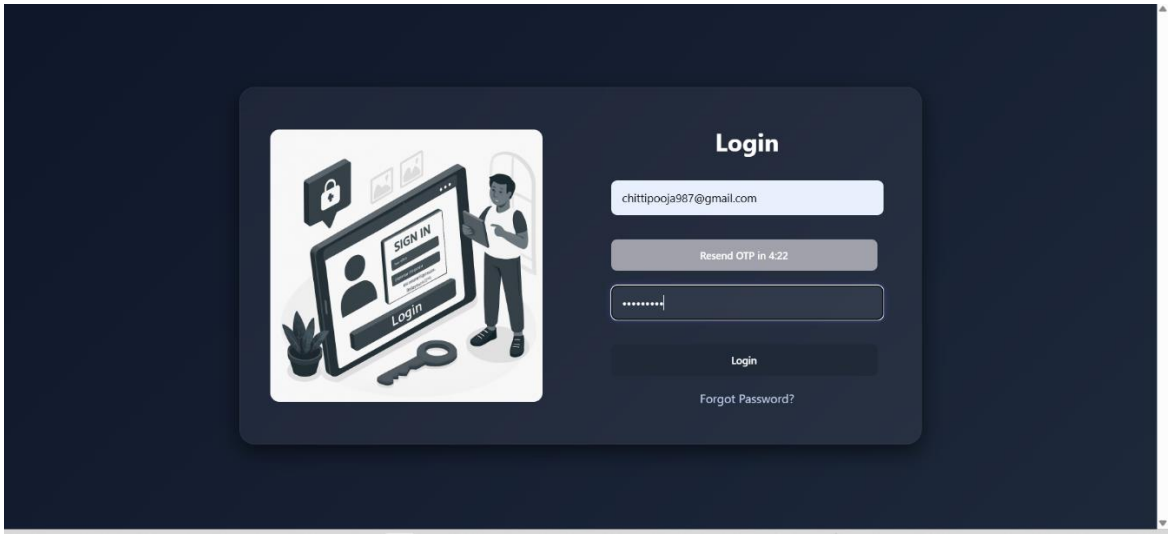
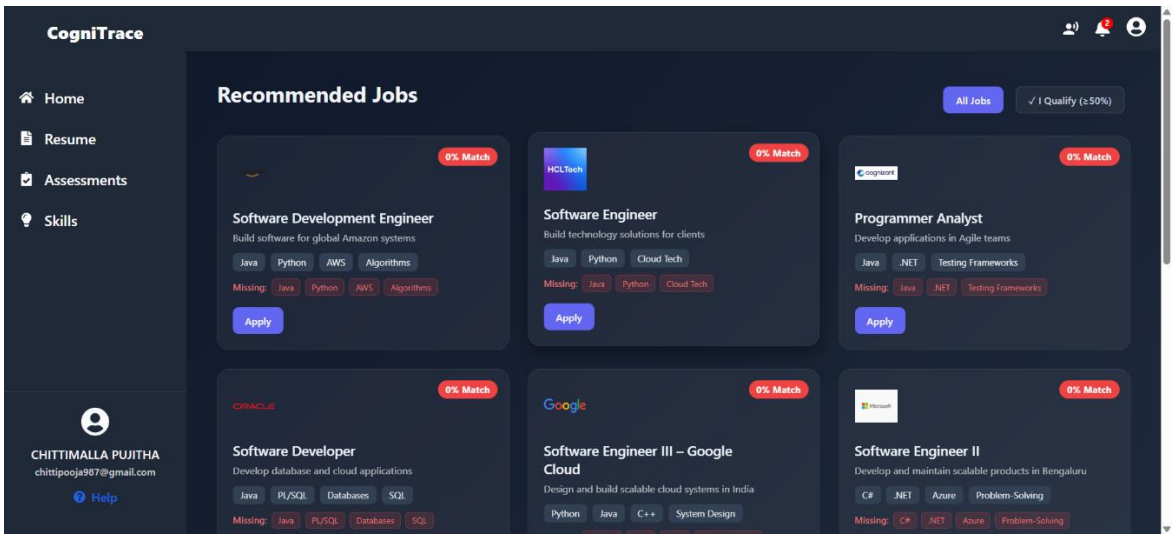
Resend OTP in 4:45
748907
Verify OTP
Forgot Password?


localhost:62216 says
OTP verified successfully.
OK
Login
chittipooja987@gmail.com
Resend OTP in 4:37
748907
Verifying...
Forgot Password?


Login
chittipooja987@gmail.com
Resend OTP in 4:22
Login
Forgot Password?
SIGN IN
Login


CogniTrace
Home
Resume
Assessments
Skills
Recommended Jobs
All Jobs
✓ I Qualify (≥50%)
0% Match
Software Development Engineer
Build software for global Amazon systems
Java Python AWS Algorithms
Missing: Java Python AWS Algorithms
Apply
Software Engineer
Build technology solutions for clients
Java Python Cloud Tech
Missing: Java Python Cloud Tech
Apply
Programmer Analyst
Develop applications in Agile teams
Java .NET Testing Frameworks
Missing: Java .NET Testing Frameworks
Apply
Software Developer
Develop database and cloud applications
Java PL/SQL Databases SQL
Missing: Java PL/SQL Databases SQL
Software Engineer III – Google Cloud
Design and build scalable cloud systems in India
Python Java C++ System Design
Software Engineer II
Develop and maintain scalable products in Bengaluru
C# .NET Azure Problem-Solving
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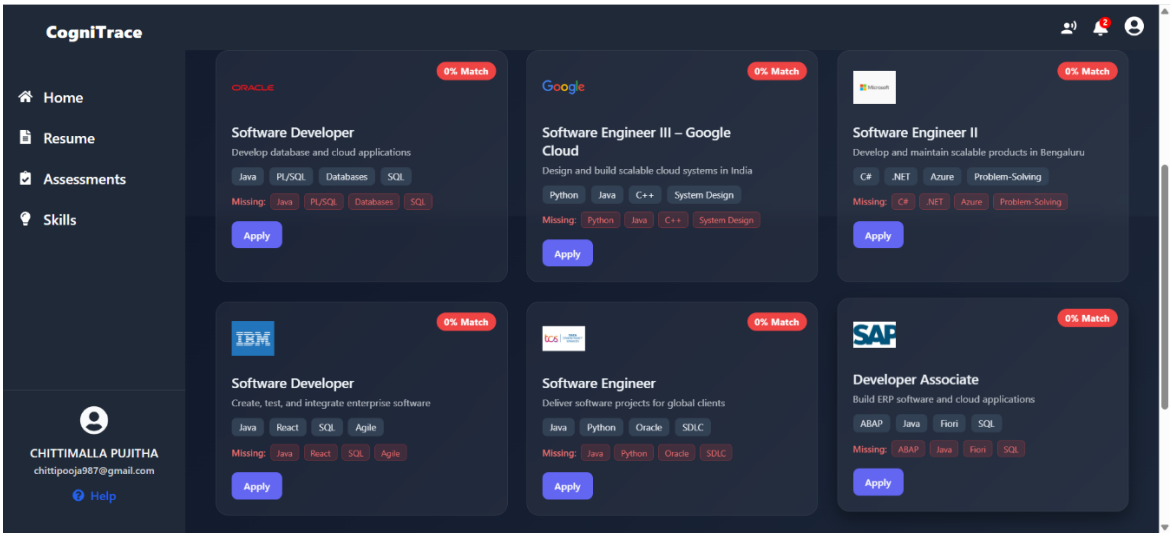
chittipooja987@gmail.com
Help


CogniTrace
Home
Resume
Assessments
Skills
0% Match
Software Developer
Develop database and cloud applications
Java PL/SQL Databases SQL
Missing: Java PL/SQL Databases SQL
Apply
Software Engineer III – Google Cloud
Design and build scalable cloud systems in India
Python Java C++ System Design
Missing: Python Java C++ System Design
Apply
Software Engineer II
Develop and maintain scalable products in Bengaluru
C# .NET Azure Problem-Solving
Missing: C# .NET Azure Problem-Solving
Apply
Software Developer
Create, test, and integrate enterprise software
Java React SQL Agile
Missing: Java React SQL Agile
Apply
Software Engineer
Deliver software projects for global clients
Java Python Oracle SDLC
Missing: Java Python Oracle SDLC
Apply
Developer Associate
Build ERP software and cloud applications
ABAP Java Fiori SQL
Missing: ABAP Java Fiori SQL
Apply
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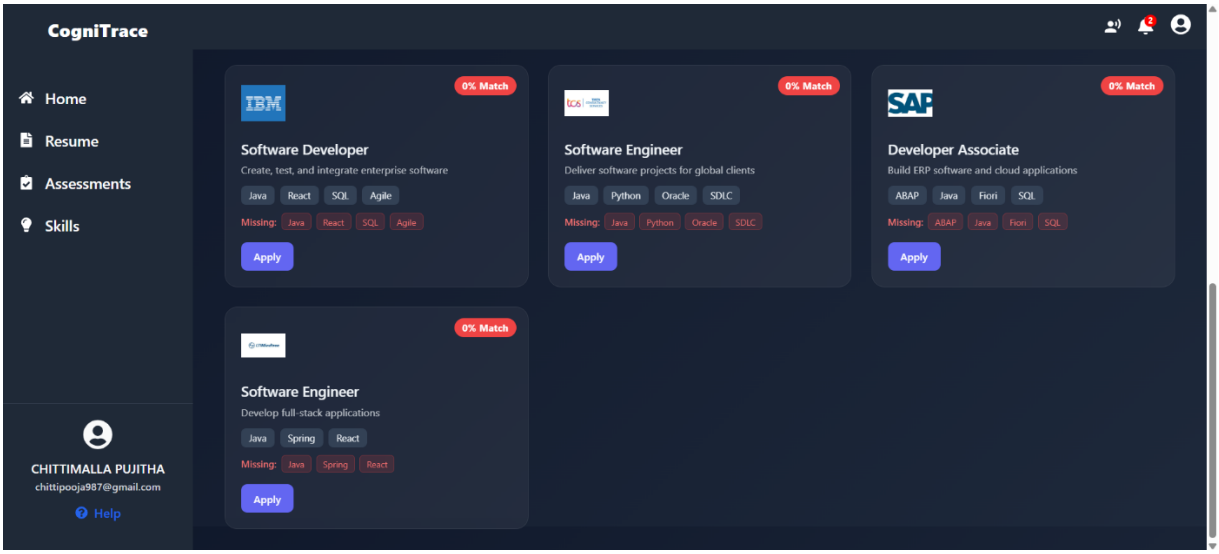
chittipooja987@gmail.com
Help


CogniTrace
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Skills
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chittipooja987@gmail.com
Help
0% Match
Software Developer
Create, test, and integrate enterprise software
Java React SQL Agile
Missing: Java React SQL Agile
Apply
0% Match
Software Engineer
Deliver software projects for global clients
Java Python Oracle SDLC
Missing: Java Python Oracle SDLC
Apply
0% Match
Developer Associate
Build ERP software and cloud applications
ABAP Java Fiori SQL
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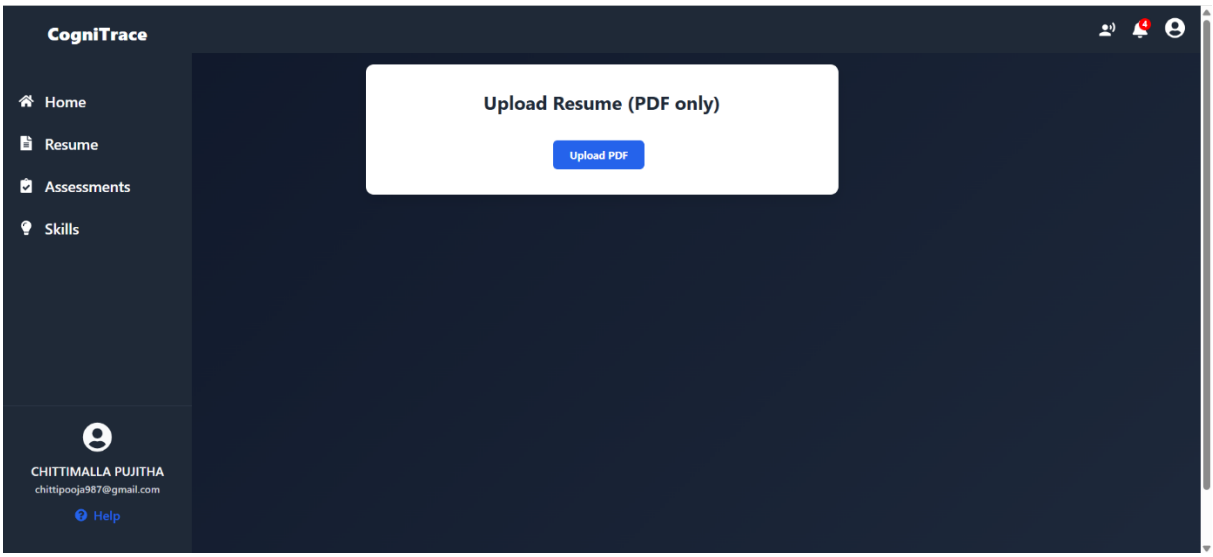
0% Match
Software Engineer
Develop full-stack applications
Java Spring React
Missing: Java Spring React
Apply


CogniTrace
Home
Resume
Assessments
Skills
CHITTIMALLA PUJITHA
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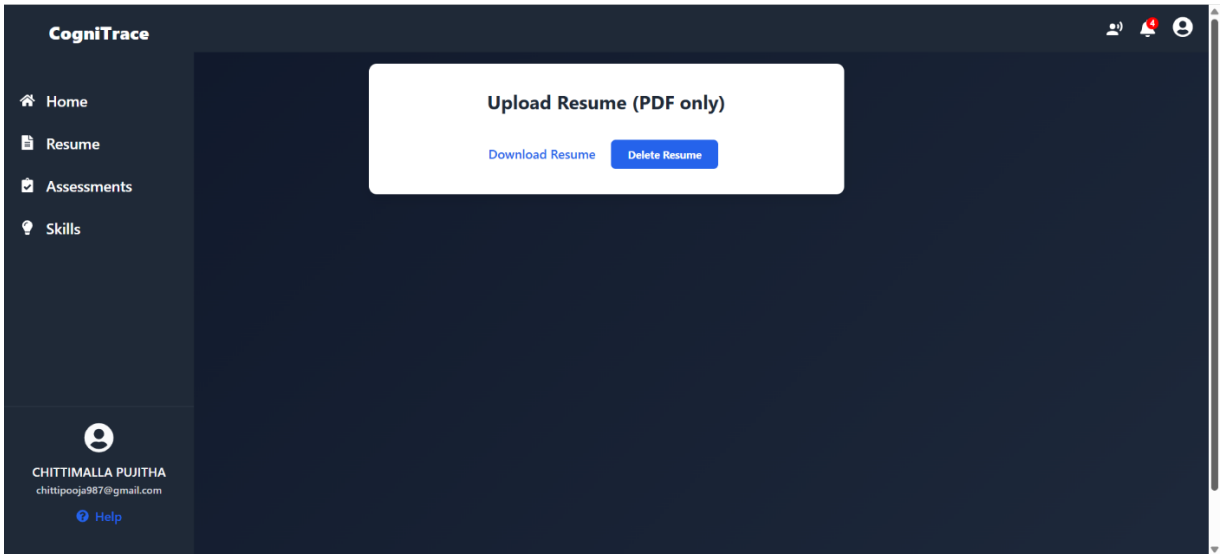
Help
Upload Resume (PDF only)
Upload PDF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Home
Resume
Assessments
Skills
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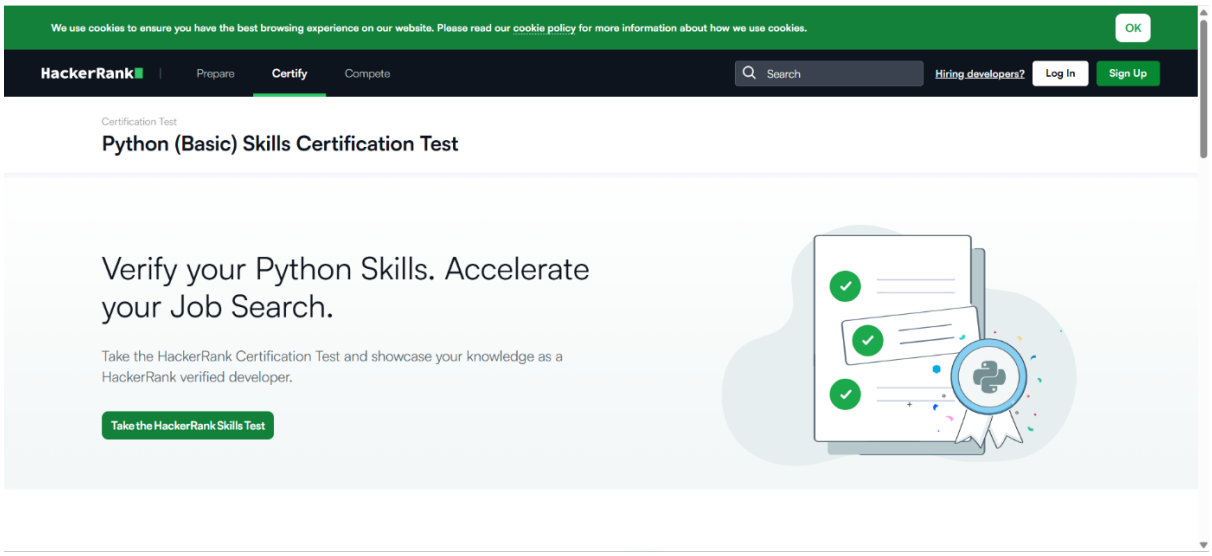
Help
Upload Resume (PDF only)
Download Resume
Delete Resume

CogniTrace
Home
Resume
Assessments
Skills
Available Assessments
Get certified in React
Attempt
Get Certified in Python
Attempt
Get Certified in Java
Attempt
Get Certified in C#
Attempt
Get Certified in SQL (Basic)
Attempt
CHITTIMALLA PUJITHA
chittipooja987@gmail.com
Help
Coudn't find your intrested Skills ?
Reach out to our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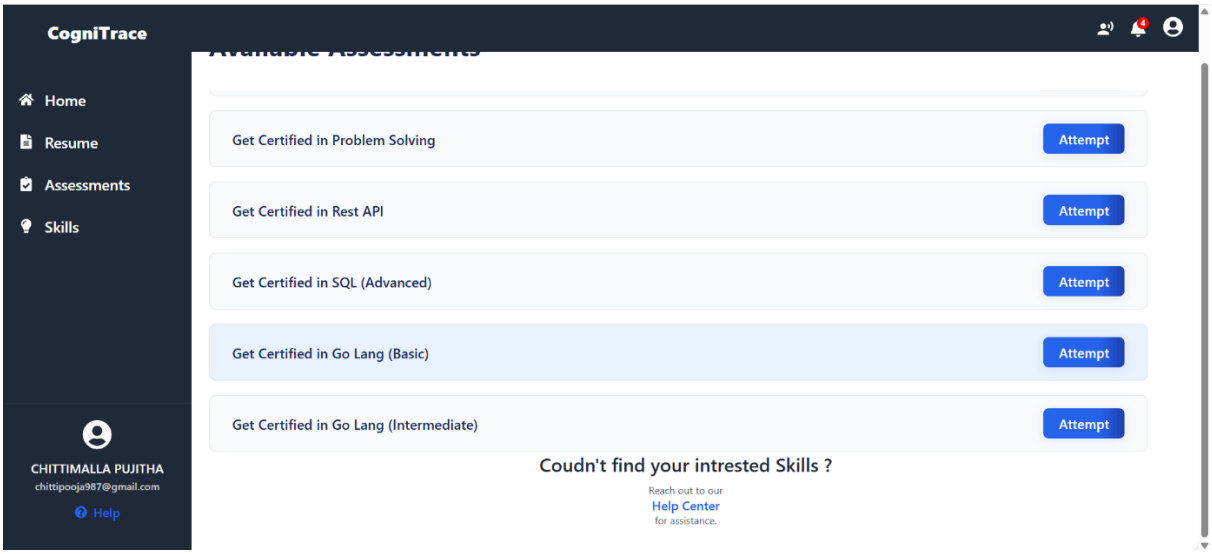
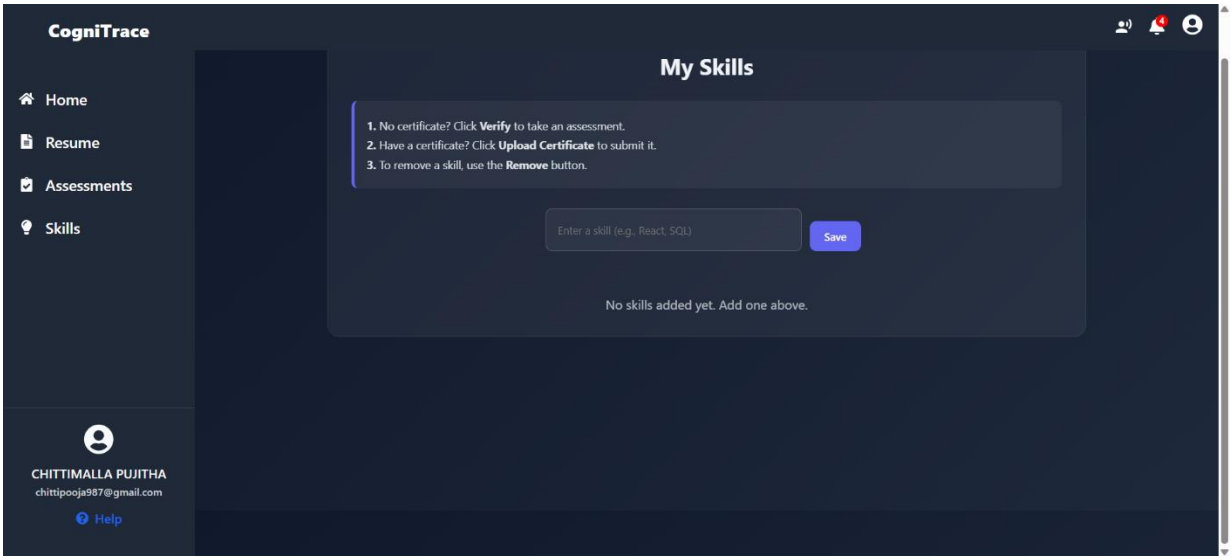
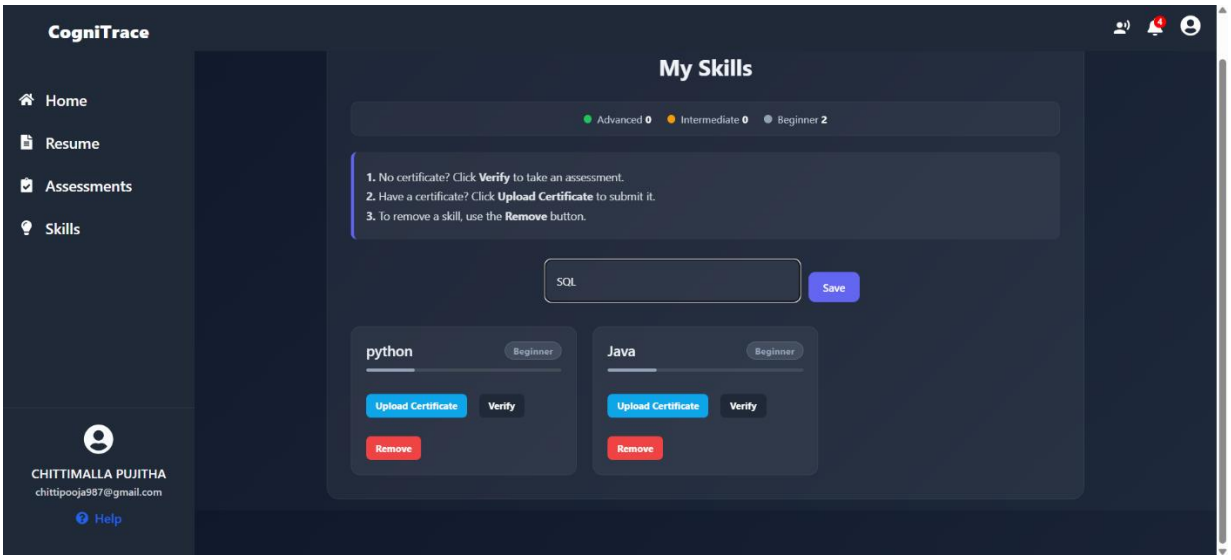
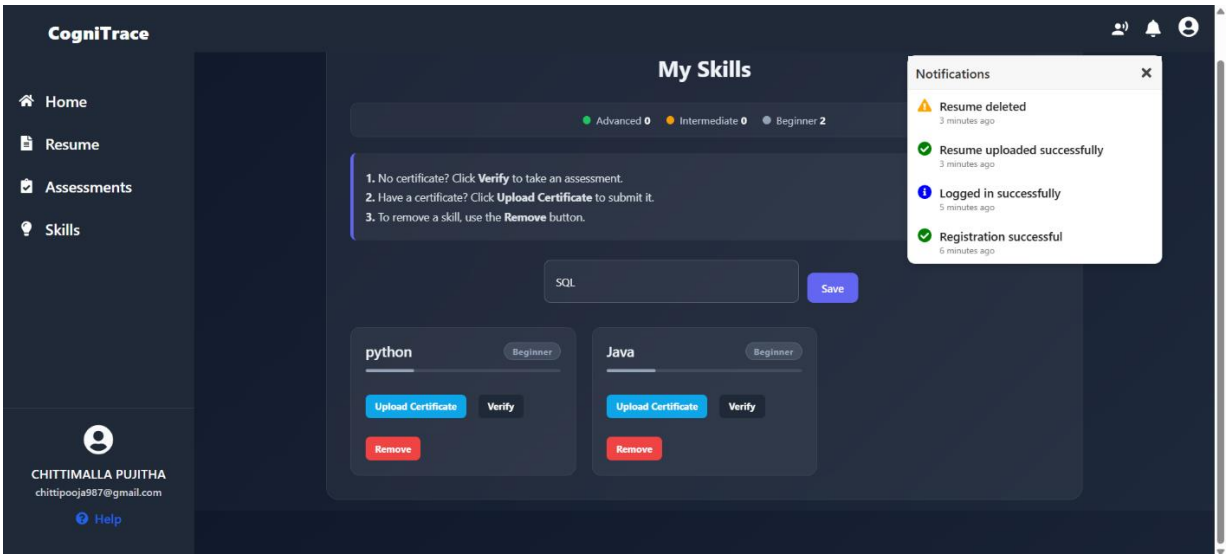
OK
HackerRank
Prepare
Certify
Compete
Search
Hiring developers?
Log In
Sign Up
Certification Test
Python (Basic) Skills Certification Test
Verify your Python Skills. Accelerate your Job Search.
Take the HackerRank Certification Test and showcase your knowledge as a HackerRank verified developer.
Take the HackerRank Skills Test


CogniTrace
Home
Resume
Assessments
Skills
Get Certified in Problem Solving
Attempt
Get Certified in Rest API
Attempt
Get Certified in SQL (Advanced)
Attempt
Get Certified in Go Lang (Basic)
Attempt
Get Certified in Go Lang (Intermediate)
Attempt
CHITTIMALLA PUJITHA
chittipooja987@gmail.com
Help
Coudn't find your intrested Skills ?
Reach out to our
Help Center
for assistance.


CogniTrace
Home
Resume
Assessments
Skills
My Skills
1. No certificate? Click Verify to take an assessment.
2. Have a certificate? Click Upload Certificate to submit it.
3. To remove a skill, use the Remove button.
Enter a skill (e.g., React, SQL)
Save
No skills added yet. Add one above.
CHITTIMALLA PUJITHA
chittipooja987@gmail.com
Help


CogniTrace
Home
Resume
Assessments
Skills
My Skills
Advanced 0
Intermediate 0
Beginner 2
1. No certificate? Click Verify to take an assessment.
2. Have a certificate? Click Upload Certificate to submit it.
3. To remove a skill, use the Remove button.
SQL
Save
python
Beginner
Upload Certificate
Verify
Remove
Java
Beginner
Upload Certificate
Verify
Remove
CHITTIMALLA PUJITHA
chittipooja987@gmail.com
Help


CogniTrace
Home
Resume
Assessments
Skills
My Skills
Advanced 0
Intermediate 0
Beginner 2
1. No certificate? Click Verify to take an assessment.
2. Have a certificate? Click Upload Certificate to submit it.
3. To remove a skill, use the Remove button.
SQL
Save
python
Beginner
Upload Certificate
Verify
Remove
Java
Beginner
Upload Certificate
Verify
Remove
Notifications
Resume deleted
3 minutes ago
Resume uploaded successfully
3 minutes ago
Logged in successfully
5 minutes ago
Registration successful
6 minutes ago
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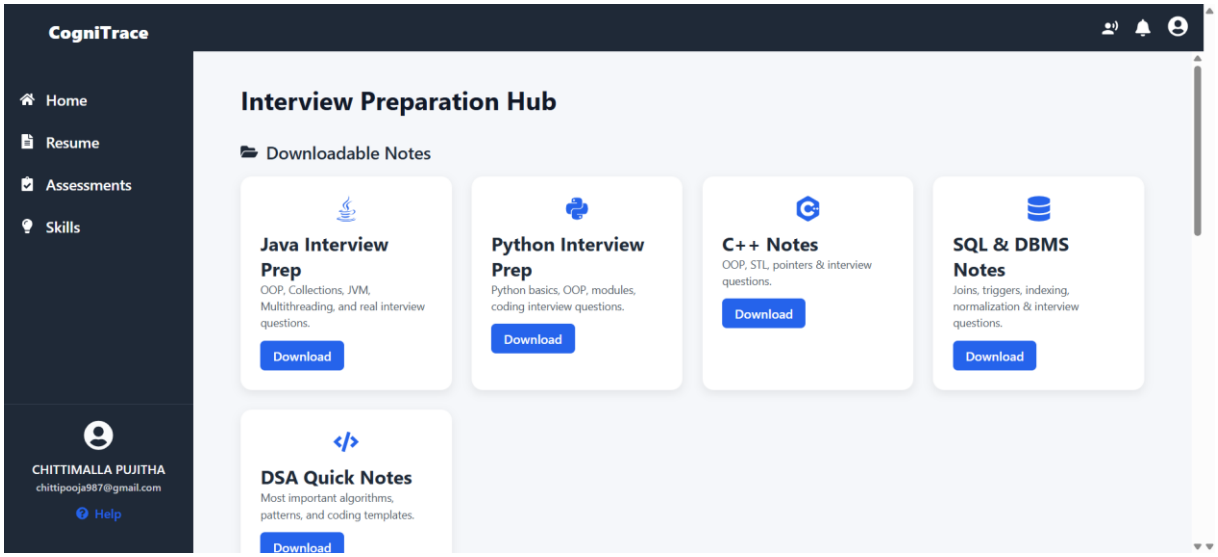
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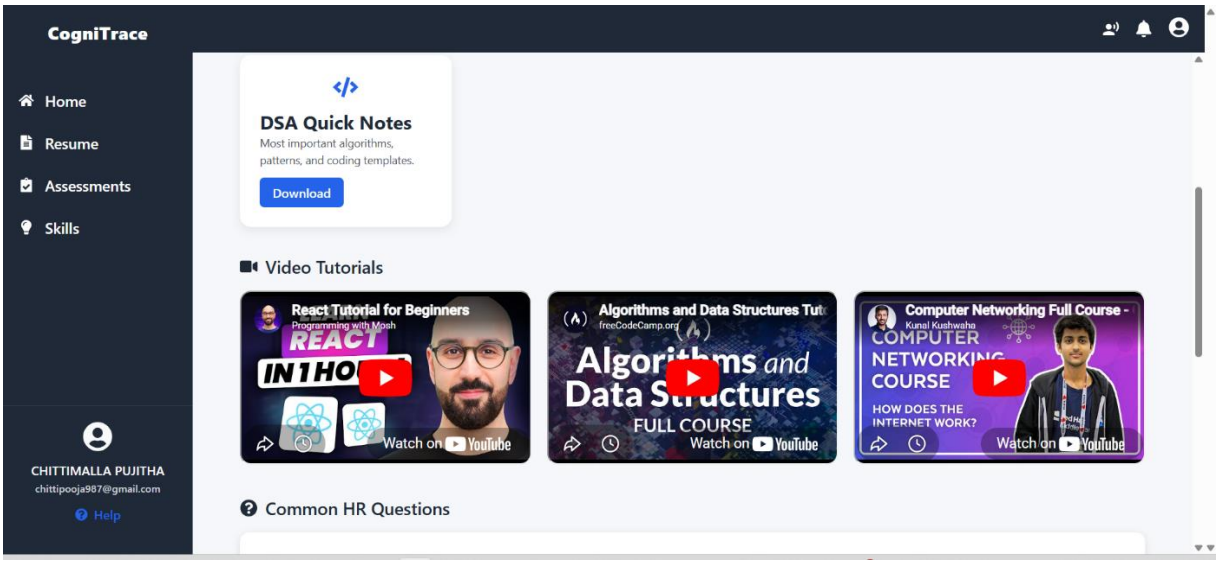
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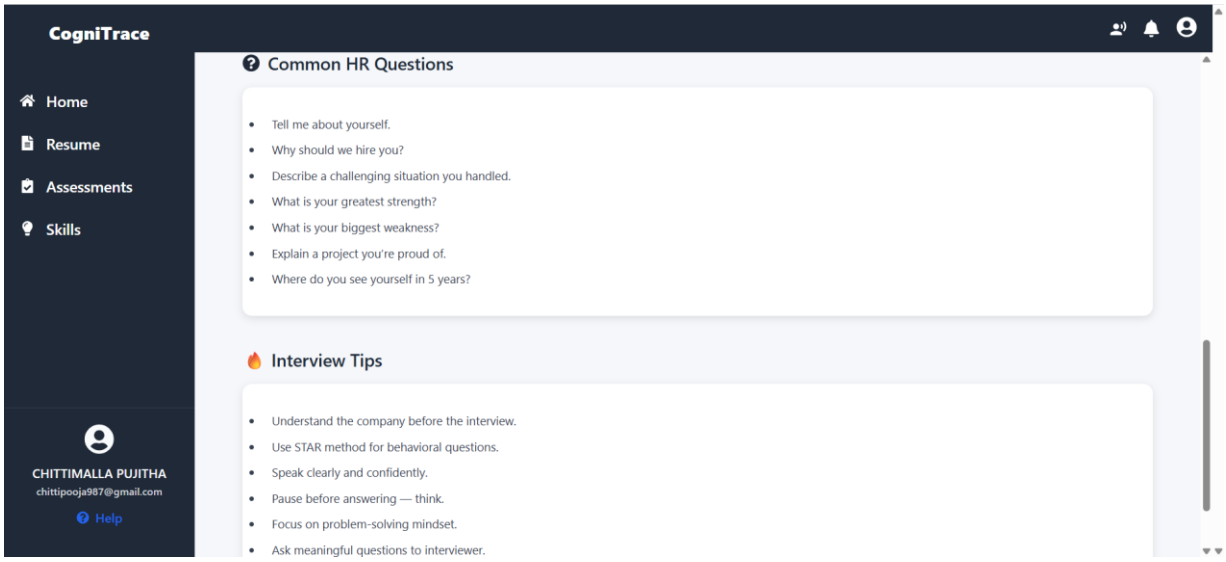
CogniTrace
Home
Resume
Assessments
Skills
Common HR Questions
Tell me about yourself.
Why should we hire you?
Describe a challenging situation you handled.
What is your greatest strength?
What is your biggest weakness?
Explain a project you're proud of.
Where do you see yourself in 5 years?
Interview Tips
Understand the company before the interview.
Use STAR method for behavioral questions.
Speak clearly and confidently.
Pause before answering — think.
Focus on problem-solving mindset.
Ask meaningful questions to interviewer.
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# 10. LEARNING OUTCOMES

**Full-Stack Web Development Skills:**

Gained hands-on experience in building a complete web application using Django for the backend and React at the frontend, integrating both layers via RESTful APIs.

**Implementation:**

Learn how to design, implement, and secure REST APIs using Django REST Framework. Including features like user authentication, skill management, file uploads, and interaction with third-party services.

**Authentication and Security:**

Developed understanding of secure user authentication using JWT tokens, OTP-based. verification, and managed file handling securely within a web application.

**Database Design and ORM Usage:**

Utilized Django's Object-Relational Mapping to design a relational database and manage it. schema, including users, skills, and associated files - resumes, certificates.

**Frontend Component Architecture:**

Built reusable and dynamic React components that handle user interactions around skills and job listings, interview preparation, resume handling, and support information.

**Having external service integration:**

Professional integration of the third-party skill validation in the form of HackerRank certifications, which allow a smooth entry into the third-party tests in an individual platform.

**File Management in Web Apps:**

Controlled those files that are uploaded by users, e.g. resumes, certificates, etc., by connecting the file storage to the user models and offering secure upload/download processes.

**Project Planning and Documentation:**

Capable of feature modularization planning, API specification, and full project documentation to maintainability and later expansion.

**User-Centered Design Approach:**

Developed the platform most specifically to serve job seekers, based on competency based job matching and interview preparation, enhancing practical and goal oriented user experiences.

**Problem-Solving and Debugging:**

Better understanding of how to troubleshoot several common problems in web development such as CORS. issues, token expiration, file upload management, and frontend-backend communication.

# 11. CONCLUSION WITH CHALLENGES

The CogniTrace project offered a lot of experience in developing a full-stack Web application, which is scalable and secure enough to handle skill-based job search and interview preparation. Among others, the most significant features that were introduced were the user authentication, skills management, resume management, and the integration with other certification platforms, thus making the features practically relevant to the users.

## Difficulties Experienced and Solution.

**OTP Redemption and OTP Generation:**

The development of a strong OTP system to validate the users was associated with a range of challenges at the beginning of the implementation, including a time constraint of delivery, expiration limits, and the validation of user inputs. It was managed by delicate backend code and user flow architecture.

**Authentication:**

The main aim of this step is to confirm the identity of a client before it is allowed to gain access to a network or any other server.

First, JWT was necessary to prevent frequent authentication errors, which were experienced following lengthy user sessions. Once turned into JWT-based authentication, all these problems disappeared and they were enhanced with refresh token systems to provide a seamless continuity of the session in a secure manner even when used over long periods of time.

**Resume Upload Handling:**

Submission of resumes had file storage restrictions, size restrictions, and configuration of the back-end. The upload failures and ensuring reliability of document management were solved by setting up of the media settings of Django with secure file handling configuration.

The solving of these dilemmas therefore enabled the project to be shifted into a full scale platform of sound authentication, proper resource control and usability that will act as a pillar to further expansion.

# 12. FUTURE SCOPE

CogniTrace is capable of expanding in rather significant directions taking into account the dynamic nature of recruitment technology and the emerging demands of students. As suggested by the literature, the role of personalization, alternative credentials, and responsible data practices will become significantly larger, and CogniTrace will have plenty of space to grow.

One of the potential paths is the utilization of smart job-recommendation systems. The contemporary machine learning frameworks know much more about the capabilities of a student, his/her interests, and experience than can be retrieved using the basic search filters. CogniTrace has the potential to go beyond job filtering by implementing some of the more advanced features, such as personal ranking, skill-gap identification, or profile-based clustering, moving it into the more of a career guide than a simple job filtering tool.

The other significant aspect is the learning and skill-building element: with the growing credibility of MOOCs and micro-credentials, the platform may suggest courses according to a student profile in an attempt to enrich the weak areas and remain in line with the industry needs. This makes the platform a place where students do not simply search jobs but they prepare them also.

It is also possible to imagine other interactive aspects that can make it even more captivating. Inclusion of functionality like virtual career fairs, in-app chat with recruiters, scheduling of interviews or automated feedback would make it easier to both students and employers.

With more predictive and data driven technology, privacy and ethics should remain in focus. In future iterations of CogniTrace it is possible to consider explainable AI, improved transparency features, and user friendly privacy control simply displaying the usage of their data. This would create credibility and provide students with greater control over their online presence.

In the long-term, CogniTrace can expand out of campus recruiting to cover opportunities with small businesses, internships, freelancing platforms, or niche industries. The platform could also be more effective with the help of collaborations with national skills registries or government employability programs.

Through these additions, CogniTrace can become a complex network that will lead students not only through the level of acquiring new skills, but also to purposeful employment instead of simply posting them on the vacancy lists.

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